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GENETICS

A causal coding variant regulating alternative splicing of *DOC2A* at 16p.11.2 GWAS locus influences susceptibility to schizophrenia

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Although genome-wide association studies (GWASs) have identified many schizophrenia-associated variants, their biological mechanisms remain unclear. Using transcriptomic data from human brain tissues, we performed splicing quantitative trait locus (sQTL) analyses of schizophrenia-associated single-nucleotide polymorphisms and identified more than 17,000 sQTLs linked to previously unidentified splicing junctions. Functional prioritization and experimental validation highlighted the synonymous variant rs3935873 within the 16p11.2 GWAS locus strongly associated with an unannotated isoform *DOC2A*$^{\Delta}$*Val217-Pro218*. rs3935873 was significantly associated with hippocampal volume, and hippocampal overexpression of *DOC2A*$^{\Delta Val217\text{-}Pro218}$ in mice recapitulated schizophrenia-relevant behavioral deficits, phenotypes absent in *DOC2A*$^{Full\text{-}Length}$-overexpressing mice. Overexpression of both isoforms altered excitatory synaptic transmission, structural modeling revealed divergent tertiary configurations between DOC2A$^{\Delta Val217\text{-}Pro218}$ and DOC2A$^{Full\text{-}Length}$, and interactome profiling highlighted that DOC2A$^{\Delta Val217\text{-}Pro218}$ unique interactors are enriched in the myosin II complex and ankyrin binding, suggesting the acquisition of previously unknown structural and regulatory functions by DOC2A$^{\Delta Val217\text{-}Pro218}$. Our study implicates dysregulated splicing in *DOC2A* as a functional mechanism for schizophrenia genetic risk and demonstrates how unannotated isoforms can reveal disease-relevant pathways.

INTRODUCTION

Schizophrenia, one of the most several psychiatric disorders, has a ~1% lifetime prevalence across distinct populations and a heritability estimate of up to 80% (*1*). Large-scale genome-wide association studies (GWASs) conducted through international consortia have identified thousands of common risk loci, providing critical insights into the polygenic architecture of this disorder (*2*). Nevertheless, because of extensive linkage disequilibrium (LD), distinguishing truly causal variants from their correlated proxy single-nucleotide polymorphisms (SNPs) remains a major challenge, impeding mechanistic interpretation of GWAS findings (*3*). Notably, most schizophrenia-associated variants reside in noncoding regions, suggesting potential regulatory roles in gene expression, as supported by expression quantitative trait locus (eQTL) analyses and functional genomic integration (*4*). However, eQTL effects explain only a fraction of the heritability, highlighting the need to investigate alternative regulatory mechanisms, such as aberrant RNA splicing, which may further bridge the gap between genetic risk and disease pathogenesis (*5*).

While eQTLs have been extensively studied, genetic regulation of RNA splicing mediated by splicing quantitative trait loci (sQTLs) represents another critical yet underexplored layer of transcriptional control. Most multiexon genes undergo alternative splicing, generating substantial transcriptomic and proteomic diversity with profound implications for phenotypic variation (*6*–*8*). Notably, sQTLs exhibit substantial independence from eQTLs (*9*), suggesting that aberrant splicing may serve as a key mechanistic link between genetic risk and schizophrenia (*10*). Precise splicing regulation is essential for normal brain function, and disruptions in this process have been implicated in schizophrenia pathogenesis (*11*, *12*). Supporting this hypothesis, schizophrenia risk SNPs show significant enrichment in prenatal brain sQTLs (*13*), and sQTL-containing loci are overrepresented among schizophrenia GWAS signals (*14*). These findings implicate the need to systematically evaluate how disease-associated genetic variants dysregulate alternative splicing and contribute to neuronal dysfunction in schizophrenia.

Although alternative splicing in the human brain exhibits substantial complexity, our understanding of transcriptomic diversity remains constrained by incomplete annotations in existing brain transcriptome datasets (*15*). While prior schizophrenia research has predominantly focused on annotated isoforms, emerging evidence highlights the critical need to characterize unannotated splicing variants and their potential disease relevance (*16*). Junction-based RNA sequencing (RNA-seq) analyses have proven particularly powerful in detecting previously unidentified splicing events that evade conventional annotation pipelines, including exon skipping, alternative 5′/3′ splice site usage, and alternative exon inclusion. Pioneering work by Jaffe *et al.* (*17*, *18*) identified numerous moderately expressed, unannotated splice junctions in human brain transcriptomes, many representing transcripts with previously unknown exonic architectures. Notably, these studies revealed significant associations between schizophrenia risk SNPs and the expression of such previously unrecognized isoforms (*18*). Notable examples include risk variants

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at 10q24.32 influencing exon 2-3 skipping in *AS3MT* (*19*) and 2q32.1 variants modulating an alternative transcription start site in *ZNF804A* (*20*). Despite these compelling findings, the full spectrum of risk-associated splicing alterations in schizophrenia remains poorly characterized, underscoring the need for systematic investigation of how genetic variants shape the splicing landscape in disease pathogenesis.

In this study, we developed an integrative analytical pipeline (fig. S1) to systematically investigate genome-wide schizophrenia GWAS risk variants for their effects on alternative splicing. Combining sQTL analysis of unannotated brain transcriptomic junctions with computational functional predictions, we identified putative causal variants and their associated unannotated isoforms with potential pathophysiological relevance. Our approach revealed that a synonymous SNP in the 16p11.2 risk locus modulates *DOC2A* alternative splicing, generating a truncated isoform that induces schizophrenia-relevant behavioral and synaptic deficits in murine models. Structural and interactome analyses demonstrated that this spliced isoform exhibits distinct conformational properties and protein-binding partners compared to full-length double C2-like domain-containing protein α (DOC2A), unveiling previously unknown molecular mechanisms underlying synaptic dysfunction in schizophrenia pathogenesis.

RESULTS

Genome-wide sQTL analysis identifies abundant schizophrenia risk SNPs associated with mRNA levels of previously unannotated splice junctions

Using GWAS summary statistics from the PGC3 (Psychiatric Genomics Consortium Phase 3) schizophrenia meta-analysis (53,386 cases and 77,258 controls), we identified 20,457 SNPs surpassing genome-wide significance ($P \leq 5.00 \times 10^{-8}$) (*2*). To mitigate confounding effects arising from complex LD patterns, we excluded SNPs within the extended major histocompatibility complex region (hg19, chr6: 25 to 35 Mb), retaining 14,463 schizophrenia risk SNPs (table S1) for subsequent analyses.

Quantifying alternative splicing through RNA-seq junction calling provides a robust approach for isoform-level transcriptome profiling. Junctions with shifted exon boundaries or with multiexon skips reflect alternative splicing events (referred to as "splice junctions") (*21*). Leveraging postmortem dorsolateral prefrontal cortex (DLPFC) RNA-seq data from two independent cohorts (BrainSeq dataset, $N = 412$ individuals; CommonMind Consortium dataset, $N = 285$ individuals) (*18*, *22*), we systematically analyzed previously unannotated splice junctions, categorizing them into "alternative exonic boundaries" and "exon skipping" events. We removed the splice junctions from the extended major histocompatibility complex region (hg19, chr6: 25 to 35 Mb) because of ambiguous short-read alignment in this highly polymorphic locus and examined the associations of remaining splicing junctions with 14,463 schizophrenia risk SNPs.

Our sQTL analysis revealed 10,575 statistically significant associations for "alternative exonic boundaries" junctions in the BrainSeq DLPFC cohort (Bonferroni-corrected $P < 0.05$; Fig. 1A), with 78.6% ($N = 8318$) demonstrating concordant allelic effects and nominal significance ($P < 0.05$) in the CommonMind Consortium replication dataset (table S2). Similarly, we identified 11,956 robust sQTLs associated with "exon skipping" junctions in BrainSeq DLPFC (Bonferroni-corrected $P < 0.05$; fig. S2), of which 73.5% ($N = 8789$) exhibited consistent effect directions and nominal significance in the independent CommonMind cohort ($P < 0.05$; table S3), demonstrating substantial cross-dataset reproducibility for both splicing event categories.

Functional predictions of schizophrenia risk loci highlight variants regulating alternative splicing

Using SpliceAI (*23*), a deep neural network that predicts splice-altering variants from pre-mRNA sequences, along with complementary tools (dbscSNV, GeneSplicer, MaxEntScan, and SpliceRegion) for cross-validation, we analyzed 14,463 schizophrenia-associated SNPs for potential splicing effects. Among these, 33 SNPs exhibited a high-confidence SpliceAI score (>0.2, indicating high recall), including 7 with scores >0.5 (recommended threshold) and 3 exceeding 0.8 (high precision) (table S4). Notably, the synonymous SNP rs3935873 (at the 16p11.2 GWAS locus) displayed the highest prediction delta score (0.87) among all tested SNPs and was computationally predicted to disrupt *DOC2A* alternative splicing (Fig. 1B).

Among the seven high-confidence splicing-modulating SNPs (SpliceAI score >0.5), rs3935873 exhibited the strongest association with an alternative exonic boundary splice junction in *DOC2A* (hg19, chr16: 30,018,412 to 30,018,499), showing robust sQTL effects in both BrainSeq DLPFC ($P = 2.25 \times 10^{-21}$) and CommonMind Consortium ($P = 6.48 \times 10^{-9}$). Similarly, rs133306 was linked to an exon skipping event in *CYP2D7P* (chr22: 42,537,350 to 42,537,877; BrainSeq, $P = 9.05 \times 10^{-7}$), although replication in CommonMind Consortium was nonsignificant ($P = 0.225$). Notably, rs2590846 influenced three previously unidentified splice variants: an alternative exonic boundary in *GNL3* (chr3: 52,720,072 to 52,720,784; BrainSeq, $P = 9.56 \times 10^{-6}$; CommonMind, $P = 1.16 \times 10^{-10}$) and two exon skipping events in *NEK4* (chr3: 52,797,641 to 52,800,193; BrainSeq, $P = 4.18 \times 10^{-10}$; CommonMind, $P = 1.80 \times$ $P = 5.18 \times 10^{-4}$; CommonMind, $P = 4.52 \times 10^{-3}$). In addition, rs2082068 modulated multiple unannotated junctions across *LRRC37A*, *LRRC37A4P*, *KANSL1*, and *MAPT*, while rs12764049 was associated with an exon skipping event in *AS3MT* (chr10: 104,629,351 to 104,632,204; BrainSeq, $P = 3.82 \times 10^{-8}$; CommonMind, $P = 3.22 \times 10^{-5}$), corresponding to the unannotated human-specific spliced isoform $AS3MT^{d2d3}$ (*19*, *24*). The remaining two SNPs (rs11692435 and rs3745474) showed no significant associations with unannotated splice junctions.

Convergent evidence prioritizes a coding variant rs3935873 regulating alternative splicing of *DOC2A* at the 16p11.2 GWAS locus

We then measured the genomic distance between risk SNPs and their corresponding sQTL junctions, hypothesizing that proximal variants are more likely to directly influence splice site usage. Notably, we identified 10 high-confidence SNP-junction pairs (<100 base pairs apart) spanning eight schizophrenia-relevant genes (*DOC2A*, *MAPT*, *KANSL1*, *IMMP2L*, *GNL3*, *NEK4*, *MAD1L1*, and *AS3MT*; Fig. 1C). A prominent example is rs3935873, a risk variant (chr16: 30,018,500) in *DOC2A*'s exon 6, positioned adjacent to an sQTL junction (chr16: 30,018,412 to 30,018,499) linked to a truncated isoform, $DOC2A^{\Delta Val217\text{-}Pro218}$, lacking six base pairs compared to the publicly annotated full-length transcript ($DOC2A^{Full\text{-}Length}$) that contains the known exon 6-7 junction (chr16: 30,018,412 to 30,018,493) (Fig. 2A). In BrainSeq DLPFC tissue, rs3935873 is markedly associated with the expression of $DOC2A^{\Delta Val217\text{-}Pro218}$ isoform ($P = 2.25 \times 10^{-21}$), and such sQTL associations were also observed in the BrainSeq

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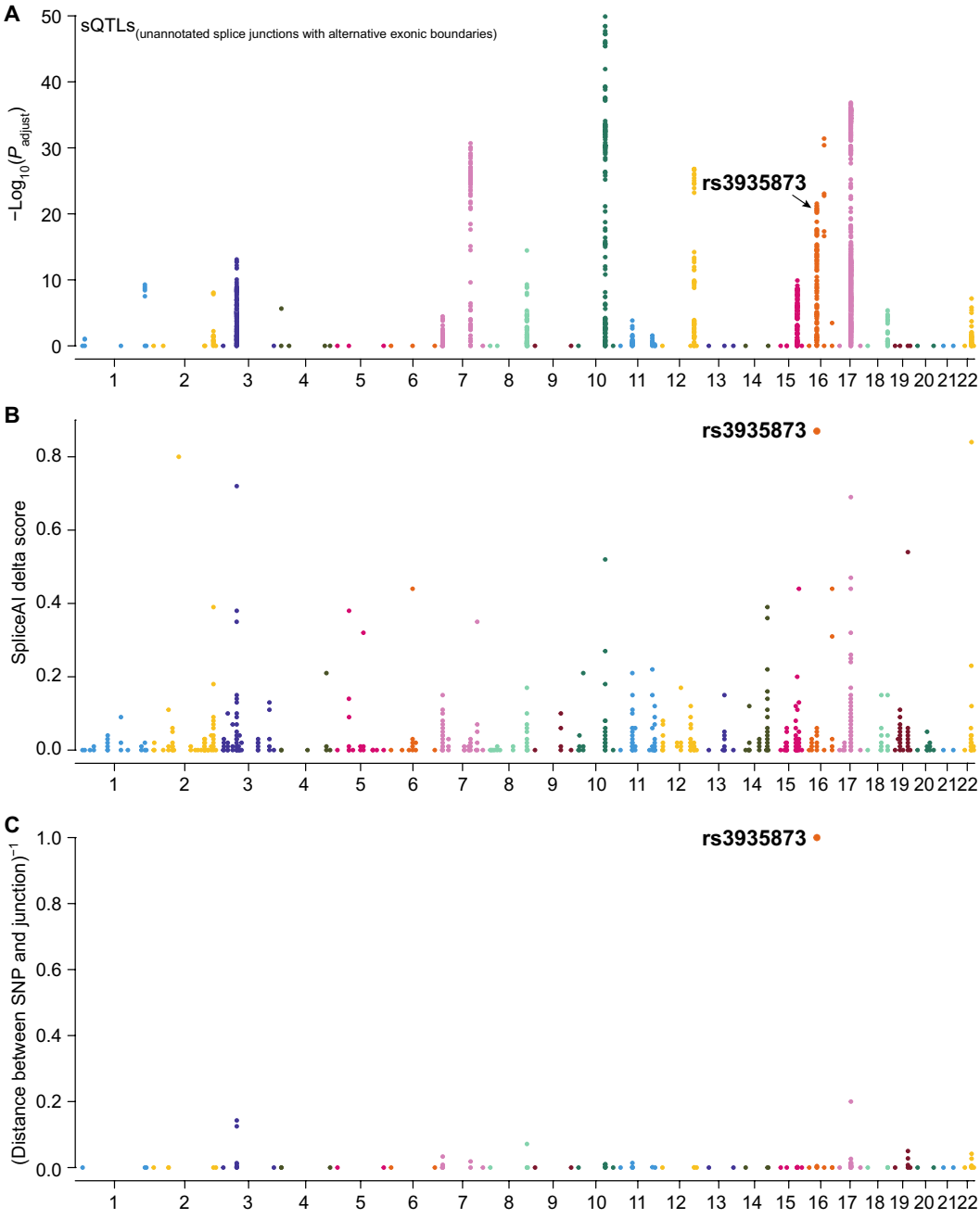

Fig. 1. Genome-wide screen of schizophrenia risk SNPs for their relevance to alternative splicing. (**A**) sQTL associations of schizophrenia risk SNPs with mRNA expression of "alternative exonic boundaries" splice junctions in the BrainSeq dataset. (**B**) Functional prediction of schizophrenia risk SNPs using SpliceAI. The higher delta score indicates a higher possibility of being functional. (**C**) Distance between the schizophrenia risk SNPs and their sQTL junctions on the genome.

hippocampus samples ($P = 0.001$) (Fig. 2B). By contrast, rs3935873 was not associated with the expression of $DOC2A^{Full\text{-}Length}$ isoform in either DLPFC ($P = 0.177$) or hippocampus ($P = 0.533$). Furthermore, rs3935873 genotypes strongly predicted the relative abundance of $DOC2A^{\Delta Val217\text{-}Pro218}$ versus $DOC2A^{Full\text{-}Length}$ (DLPFC: $P < 0.0001$; hippocampus: $P = 0.0035$; Fig. 2B), underscoring its role in modulating isoform balance. We also assessed the eQTL association between rs3935873 and the gene-level expression of *DOC2A* in DLPFC and hippocampus, and neither of them were significant ($P > 0.3$; fig. S3), which is expected, as *DOC2A* encoded multiple transcripts, and $DOC2A^{Full\text{-}Length}$ is the primary isoform.

To functionally characterize the splicing regulatory effects of rs3935873, we engineered *DOC2A* minigene constructs encompassing intron 5 through exon 8 (including alternative rs3935873 alleles) and transfected them into Neuro-2a, human embryonic kidney (HEK) 293T, and SH-SY5Y cell lines. Quantitative analysis revealed a significant allele-specific effect on the expression of the alternatively spliced $DOC2A^{\Delta Val217\text{-}Pro218}$ transcript, with the rs3935873 G-allele

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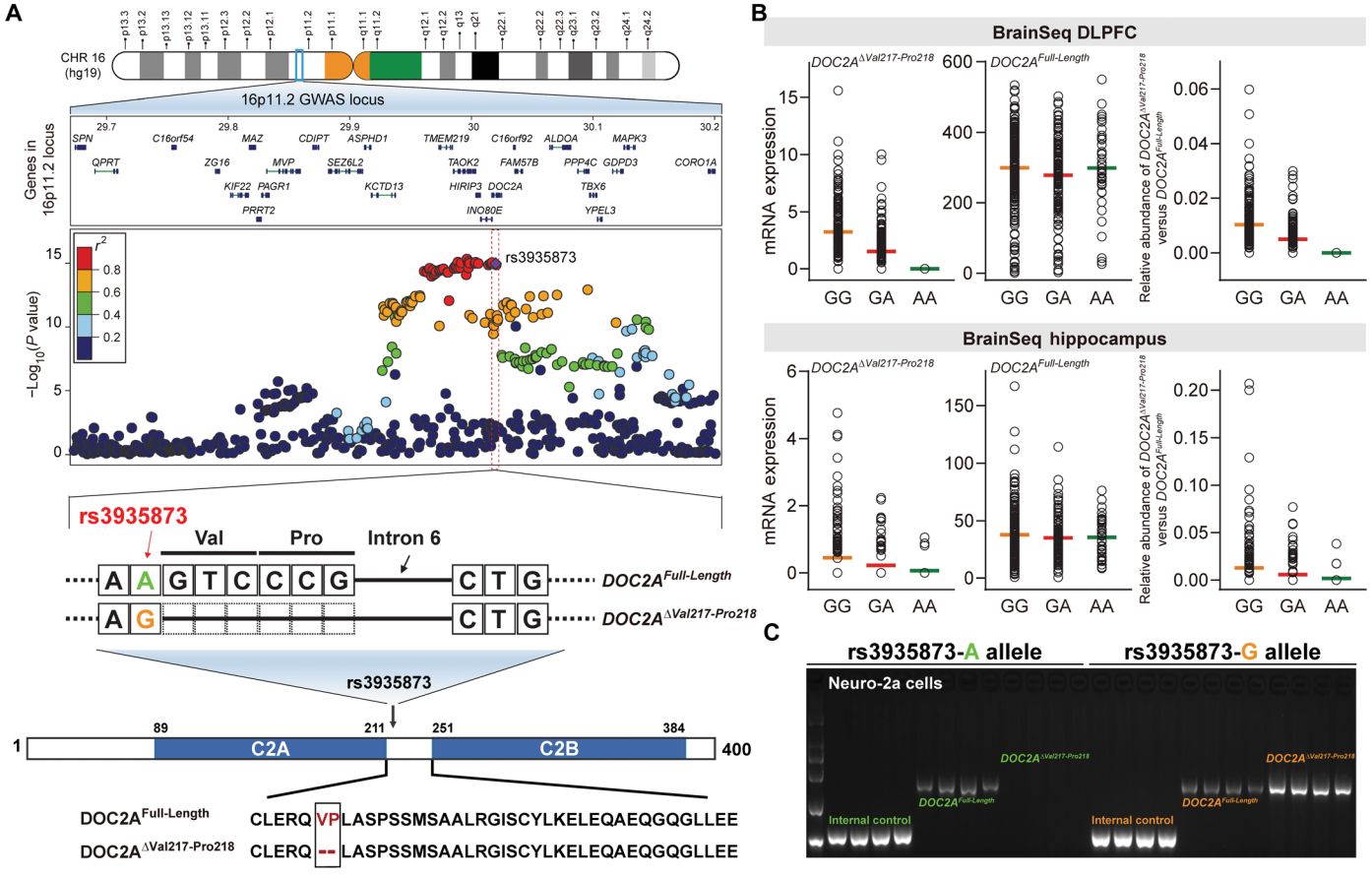

Fig. 2. Molecular characterization of rs3935873. (**A**) Genetic association of SNPs spanning the 16p11.2 GWAS locus with schizophrenia and schematic of rs3935873 locations in *DOC2A*. A physical map of the region is given and depicts known genes within the region, and the LD is defined on the basis of the SNP rs3935873. (**B**) sQTL analyses of rs3935873 with mRNA expression of *DOC2A*$^{\Delta Val217\text{-}Pro218}$ and *DOC2A*$^{Full\text{-}Length}$ and ratio (*DOC2A*$^{\Delta Val217\text{-}Pro218}$/*DOC2A*$^{Full\text{-}Length}$) in the BrainSeq DLPFC and hippocampus dataset. (**C**) Experimental design and results of in vitro minigene experiments in Neuro-2a cells. PCR results indicate that the different alleles of rs3935873 exert a significant impact on the expression of *DOC2A*$^{\Delta Val217\text{-}Pro218}$ rather than *DOC2A*$^{Full\text{-}Length}$.

showing strong positive association across all cell models. This finding corroborates our sQTL analyses in human DLPFC and hippocampal tissues. Specifically, cells carrying the GG-genotype of rs3935873 consistently had a greater abundance of *DOC2A*$^{\Delta Val217\text{-}Pro218}$ isoform compared with those carrying the AA-genotype in multiple assays [semiquantitative polymerase chain reaction (PCR) results in Neuro-2a cells are shown in Fig. 2C, and results in HEK293T and SH-SY5Y cells are shown in fig. S4], demonstrating consistent cis-regulatory effects regardless of cellular context. Cross-species transcriptomic analysis further revealed that the *DOC2A*$^{\Delta Val217\text{-}Pro218}$ isoform appears to be human- or primate-specific, with no detectable expression in murine hippocampal or cortical tissues. These functional genomics data collectively establish rs3935873 as a causal genetic variant governing *DOC2A* alternative splicing.

The causal variant rs3935873 is associated with hippocampal volume

Given the well-documented alterations in brain structure and regional volumes in patients with schizophrenia and their unaffected first-degree relatives (*25*–*27*), we investigated whether these neuroanatomical changes were linked to the schizophrenia risk allele of rs3935873. Leveraging large-scale imaging-genetics data from the ENIGMA consortium (https://enigma-brain.org/enigmavis/) (*28*), we performed a PheWeb-based brainwide association analysis and identified a robust association between rs3935873 and hippocampal volume ($P = 6.44 \times 10^{-10}$, $N = 74{,}898$; Fig. 3) (*29*), with no significant effects on other neuroimaging traits (table S5). This meta-analysis combined data from the ENIGMA and CHARGE cohorts ($N = 33{,}536$), UK Biobank ($N = 36{,}095$), and the ABCD study ($N = 5267$), revealing that the schizophrenia-risk G-allele was associated with increased hippocampal volume. Given that rs3935873 regulates the alternative splicing of *DOC2A*$^{\Delta Val217\text{-}Pro218}$, these results suggest a potential role for this transcript in hippocampal biology and schizophrenia pathogenesis.

Overexpression of *DOC2A*$^{\Delta Val217\text{-}Pro218}$ in the hippocampus led to schizophrenia-like behaviors

To investigate the behavioral consequences of *DOC2A*$^{\Delta Val217\text{-}Pro218}$ overexpression, we targeted the dorsal hippocampus of adult male C57BL/6J mice (7 to 8 weeks old) using adeno-associated virus (AAV)–mediated bilateral delivery, given that this isoform is detectable in adult brain RNA-seq data, but its developmental expression profile remains uncharacterized. This brain region was selected on the basis of recent spatial transcriptomics evidence implicating schizophrenia-associated glutamatergic neurons near the dorsal

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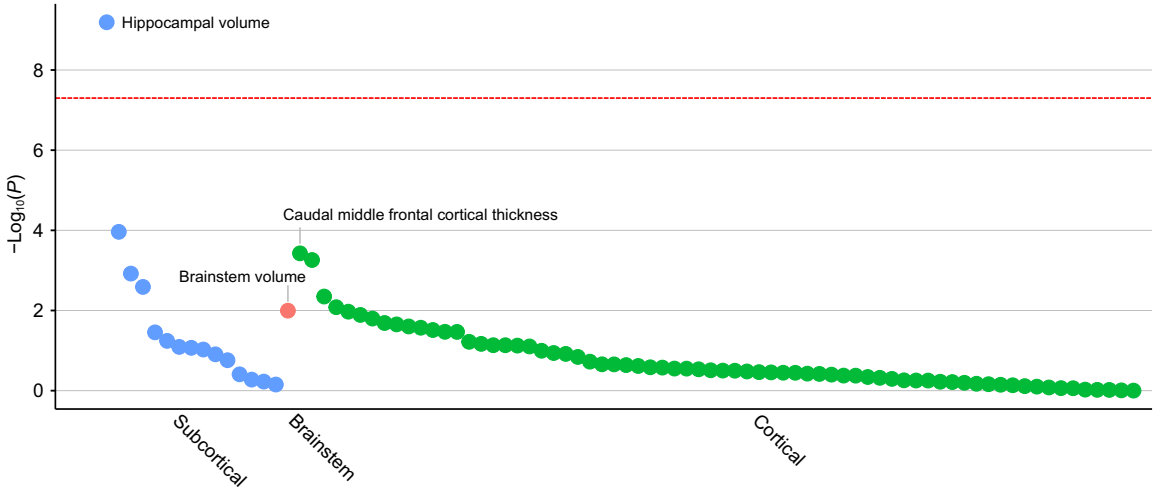

Fig. 3. Association of rs3935873 with brain imaging phenotypes. Hippocampal volume exhibited the strongest associations with rs3935873.

hippocampus (*30*), a locus functionally linked to spatial memory (*31*) and known to elicit schizophrenia-relevant phenotypes [e.g., hyperactivity and protein-protein interaction (PPI) deficits] upon dopaminergic modulation (*32*, *33*). Immunofluorescence analysis of brain slices confirmed robust overexpression of both DOC2A$^{\Delta Val217\text{-}Pro218}$ and DOC2A$^{Full\text{-}Length}$ isoforms (fig. S5). Injection sites in the hippocampus were visualized at 4 weeks postinjection (fig. S6), with no adverse effects on animal health and hippocampal volume (fig. S7). These results establish a clean model for dissecting isoform-specific behavioral effects without neuroanatomical alterations.

We used the open field test to measure the locomotor activities of the mice. There was no difference in the total traveled distance between control mice and *DOC2A*$^{Full\text{-}Length}$-overexpressing mice or *DOC2A*$^{\Delta Val217\text{-}Pro218}$-overexpressing mice [control, 13158.0 ± 487.40 mm in total distance; *DOC2A*$^{Full\text{-}Length}$, 12902.0 ± 1138.0 mm in total distance; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 13476.0 ± 1018.0 mm in total distance; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.85$, $t = 0.20$, degrees of freedom (df) = 42; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.81$, $t = 0.24$, df = 42; multiple comparisons using Fisher's least significant difference (LSD) in a one-way analysis of variance (ANOVA); Fig. 4A]. Compared with control mice, *DOC2A*$^{Full\text{-}Length}$ mice and *DOC2A*$^{\Delta Val217\text{-}Pro218}$ mice showed a significant reduction in the percentage of distance traveled in the center area (control, 14.23 ± 1.11% in distance traveled in the center area; *DOC2A*$^{Full\text{-}Length}$, 9.88 ± 0.97% in distance traveled in the center area; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 10.04 ± 1.12% in distance traveled in the center area; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.006$, $t = 2.88$, df = 42; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.008$, $t = 2.77$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4A) and a notable increase in the percentage of distance traveled in the corner area (control, 31.58 ± 1.10% in distance traveled in the corner area; *DOC2A*$^{Full\text{-}Length}$, 35.31 ± 1.20% in distance traveled in the corner area; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 35.55 ± 1.25% in distance traveled in the corner area; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.03$, $t = 2.23$, df = 42; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.02$, $t = 2.37$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4A). However, no significant differences were observed in time spent in the center area (control, 28.25 ± 4.12 s in time spent in the center area; *DOC2A*$^{Full\text{-}Length}$, 19.18 ± 2.59 s in time spent in the center area; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 22.06 ± 4.29 s in time spent in the center area; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.09$, $t = 1.71$, df = 42; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.25$, $t = 1.17$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4A) or corner area (control, 139.80 ± 8.60 s in time spent in the corner area; *DOC2A*$^{Full\text{-}Length}$, 158.80 ± 7.02 s in time spent in the corner area; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 158.60 ± 7.85 s in time spent in the corner area; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.09$, $t = 1.72$, df = 42; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.10$, $t = 1.70$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4A).

We also performed the elevated zero maze (EZM) test to evaluate anxiety levels of the mice. During 5 min of free exploration, no significant differences were observed in the total distance traveled among the three groups of mice (control, 6758.0 ± 705.2 mm in total distance; *DOC2A*$^{Full\text{-}Length}$, 8024.0 ± 723.4 mm in total distance; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 6112.0 ± 563.0 mm in total distance; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.19$, $t = 1.34$, df = 42; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.50$, $t = 0.68$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4B). However, the *DOC2A*$^{\Delta Val217\text{-}Pro218}$-overexpressing mice exhibited a notable reduction in the percentage of distance traveled in the open arms (control, 18.67 ± 3.0% distance in open arms; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 11.29 ± 2.19% distance in open arms; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.05$, $t = 2.02$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4B), while the percentage of distance traveled in the closed arms showed a trend increase, but the difference was not statistically significant (control, 79.61 ± 2.94% distance in closed arms; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 86.14 ± 2.04% distance in closed arms; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.07$, $t = 1.84$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4B) compared to the control mice. In contrast, *DOC2A*$^{Full\text{-}Length}$ mice displayed no significant behavioral differences from the control mice (control, 18.67 ± 3.0% distance in open arms; *DOC2A*$^{Full\text{-}Length}$, 18.52 ± 2.53% distance in open arms; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 1.00$, $t = 0.04$, df = 42; control, 79.61 ± 2.94% distance in closed arms; *DOC2A*$^{Full\text{-}Length}$, 79.56 ± 2.44% distance in closed

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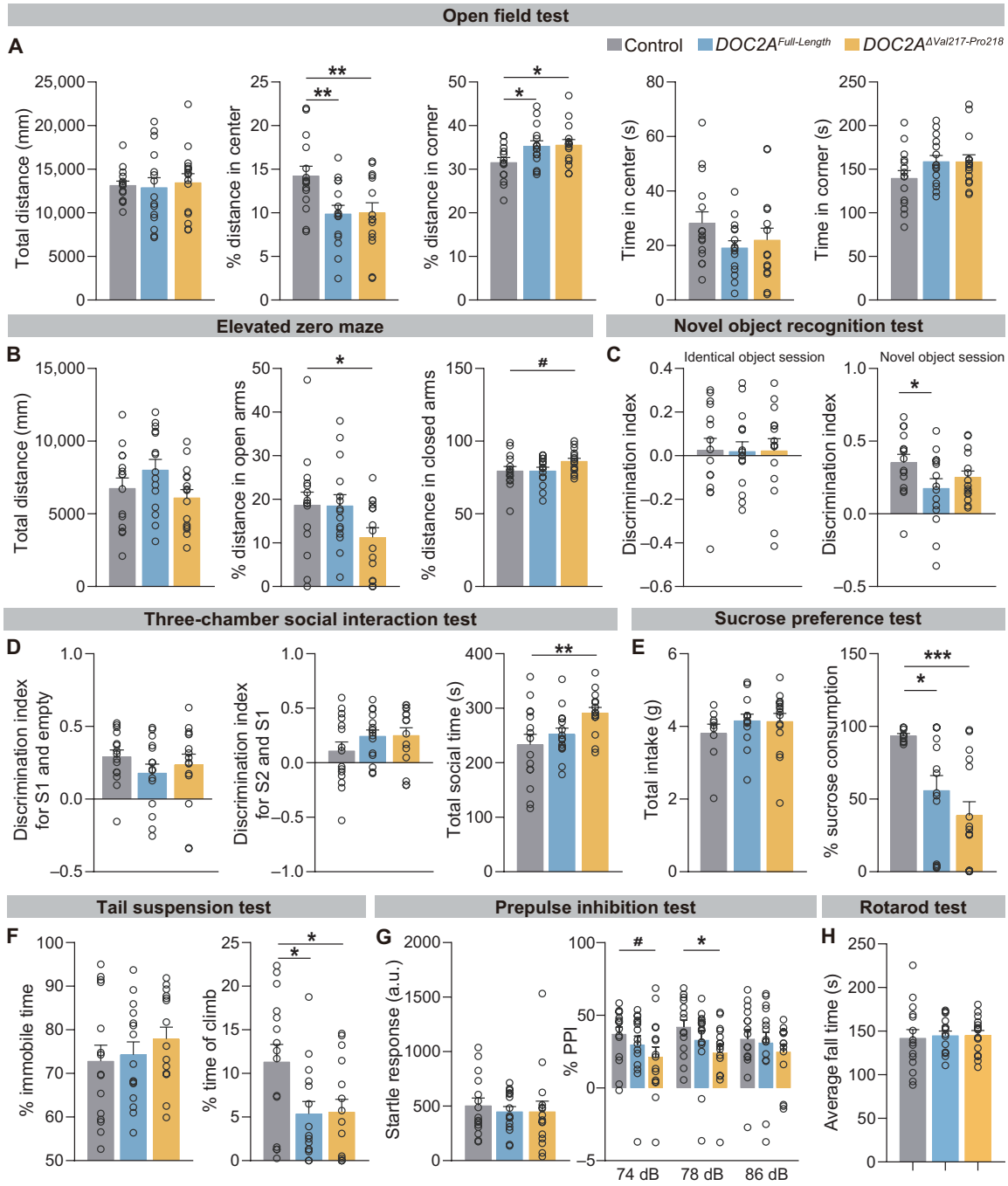

Fig. 4. Overexpression of *DOC2A*$^{\Delta Val217\text{-}Pro218}$ in the hippocampus led to schizophrenia-like behaviors. (**A**) Open field test. Left, total distance traveled; left middle, the percentage of distance traveled in the center area; middle, the percentage of distance in the corner area; right middle, time spent in the center area; right, time spent in the corner area. $N = 15$ in each group. (**B**) EZM. Left, total distance traveled; middle, the percentage of distance traveled in open arms; right, the percentage of distance traveled in closed arms. $N = 15$ in each group. (**C**) Novel object recognition test. Left, discrimination index of identical object session; right, discrimination index of novel object session. $N = 15$ in each group. (**D**) Three-chamber social interaction test. Left, discrimination index of the social stage; middle, discrimination index of the social novelty stage; right, total social time in the social novelty stage. $N = 15$ in each group. (**E**) SPT. Left, total intake; right, the percentage of sucrose consumption. $N \geq 9$ in each group. (**F**) TST. Left, the percentage of immobile time; right, the percentage of climb time. $N = 15$ in each group. (**G**) Prepulse inhibition test. Left, the startle response; right, the percentage of PPI of the startle response on 74-, 78-, and 84-dB prepulse intensities. $N = 15$ in each group. a.u., arbitrary units. (**H**) Rotarod test. $N = 15$ in each group. #$P < 0.1$, *$P < 0.05$, **$P < 0.01$, and ***$P < 0.001$. Error bars indicate the SEM.

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arms; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 1.00$, $t = 0.01$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4B). These findings indicate that the *DOC2A*$^{\Delta Val217\text{-}Pro218}$-overexpressing mice exhibit elevated anxiety-like behaviors.

In the behavioral tests for the assessment of recognition memory (*34*), the discrimination index showed no significant difference during the identical object session in either *DOC2A*$^{Full\text{-}Length}$ mice or *DOC2A*$^{\Delta Val217\text{-}Pro218}$ mice (control, 0.03 ± 0.05 discrimination index; *DOC2A*$^{Full\text{-}Length}$, 0.02 ± 0.04 discrimination index; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 0.02 ± 0.05 discrimination index; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.94$, $t = 0.08$, df = 42; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.98$, $t = 0.03$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4C). Nevertheless, *DOC2A*$^{Full\text{-}Length}$ mice had a lower discrimination index (control, 0.35 ± 0.06 discrimination index; *DOC2A*$^{Full\text{-}Length}$, 0.18 ± 0.07 discrimination index; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.03$, $t = 2.26$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4C), and *DOC2A*$^{\Delta Val217\text{-}Pro218}$ had a normal recognition ability (control, 0.35 ± 0.06 discrimination index; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 0.25 ± 0.04 discrimination index; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.20$, $t = 1.29$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4C) in the novel object session. In the Y maze test, there were no statistical differences in the time spent in the novel arm or numbers of entries to the novel arm among the control mice, *DOC2A*$^{\Delta Val217\text{-}Pro218}$-overexpressing mice, and *DOC2A*$^{Full\text{-}Length}$-overexpressing mice (fig. S8A), and mice in all three groups showed similar spontaneous alterations (fig. S8B). In the three-chamber social interaction test, there were no differences in the discrimination indices among all groups in the social stage or social novel stage (Fig. 4D). However, the total interaction social time with familiar and stranger mice was significantly higher in *DOC2A*$^{\Delta Val217\text{-}Pro218}$ mice than in control mice (control, 233.3 ± 18.70 s in total social time; *DOC2A*$^{Full\text{-}Length}$, 252.2 ± 11.35 s in total social time; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 291.60 ± 10.01 s in total social time; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.34$, $t = 1.0$, df = 42; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.005$, $t = 3.0$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4D), indicating normal social novelty preference and increased social exploration in *DOC2A*$^{\Delta Val217\text{-}Pro218}$-overexpressing mice. Together, *DOC2A*$^{Full\text{-}Length}$-overexpressing mice exhibited recognition memory deficits, and *DOC2A*$^{\Delta Val217\text{-}Pro218}$ mice had normal recognition memory. These results indicate that *DOC2A*$^{\Delta Val217\text{-}Pro218}$ mice exhibit distinct cognitive and social functions compared with *DOC2A*$^{Full\text{-}Length}$ mice.

Schizophrenia is characterized by prominent negative symptoms including anhedonia, affective blunting, avolition, alogia, and social isolation. Behavioral phenotyping revealed significant anhedonic-like behavior in both *DOC2A* isoform–overexpressing murine models compared to controls, as evidenced by reduced sucrose preference in the sucrose preference test (SPT) (control, 93.56 ± 1.50% sucrose preference; *DOC2A*$^{Full\text{-}Length}$, 55.99 ± 10.24% sucrose preference; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 38.66 ± 9.58% sucrose preference; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.01$, $t = 2.65$, df = 35; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.0004$, $t = 3.93$, df = 35; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4E). Consistent with these findings, tail suspension test (TST) analyses demonstrated markedly decreased climbing time in *DOC2A* isoform–overexpressing mice (control, 11.32 ± 2.0% in time of climb; *DOC2A*$^{Full\text{-}Length}$, 5.37 ± 1.40% in time of climb; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 5.57 ± 1.46% in time of climb; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.01$, $t = 2.56$, df = 42; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.02$, $t = 2.48$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4F), suggesting impaired stress-coping mechanisms. These behavioral deficits recapitulate core negative symptom domains observed in patients with schizophrenia, with the alternatively spliced *DOC2A*$^{\Delta Val217\text{-}Pro218}$ isoform showing particularly pronounced effects on reward processing.

PPI of the startle reflex (*31*), a well-established measure of sensorimotor gating deficits in schizophrenia (*35*, *36*), assesses the attenuation of a startle response (pulse) when preceded by a weaker stimulus (prepulse). Impaired PPI reflects disrupted early-stage information processing, a hallmark feature observed in patients with schizophrenia (*37*, *38*). In our PPI paradigm, combining an auditory startle stimulus (120 dB) with prepulse intensities (74, 78, and 86 dB), we evaluated the effects of *DOC2A*$^{\Delta Val217\text{-}Pro218}$ and *DOC2A*$^{Full\text{-}Length}$ overexpression on sensorimotor gating. Baseline startle responses did not differ between groups (control, 503.4 ± 71.02 startle responses; *DOC2A*$^{Full\text{-}Length}$, 448.4 ± 47.94 startle responses; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 449.7 ± 96.20 startle responses; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.60$, $t = 0.52$, df = 42; DOC2A$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.61$, $t = 0.51$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4G), confirming intact acoustic startle reflex and hearing function. Notably, *DOC2A*$^{\Delta Val217\text{-}Pro218}$ mice exhibited reduced PPI at 74 dB (control, 37.01 ± 4.78% PPI; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 21.28 ± 6.99% PPI; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.07$, $t = 1.84$, df = 126; multiple comparisons using Fisher's LSD in a two-way ANOVA; Fig. 4G) and significantly impaired gating at 78 dB (control, 41.91 ± 4.94% PPI; DOC2A$^{\Delta Val217\text{-}Pro218}$, 24.23 ± 5.73% PPI; DOC2A$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.04$, $t = 2.06$, df = 126; multiple comparisons using Fisher's LSD in a two-way ANOVA; Fig. 4G), suggesting selective sensorimotor gating deficits. In contrast, *DOC2A*$^{Full\text{-}Length}$ mice displayed normal PPI, indicating isoform-specific effects on sensorimotor processing. The motor function (rotarod test: control, 142.0 ± 9.85 s in average fall time; *DOC2A*$^{Full\text{-}Length}$, 144.8 ± 5.24 s in average fall time; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 145.1 ± 5.59 s in average fall time; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.78$, $t = 0.28$, df = 42; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.76$, $t = 0.30$, df = 42; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 4H) showed no obvious difference among them.

Considering the role of the prefrontal cortex (PFC) in schizophrenia, we bilaterally delivered AAV vectors encoding either *DOC2A*$^{\Delta Val217\text{-}Pro218}$ or *DOC2A*$^{Full\text{-}Length}$ into the PFC of wild-type C57BL/6J male mice (fig. S9) to investigate region-specific effects of *DOC2A* isoform expression. Comprehensive behavioral phenotyping revealed no significant alterations in locomotor activity (open field test), anxiety-like behaviors (EZM), cognitive function (novel object recognition), sociability (three-chamber social interaction), anhedonia (sucrose preference), depressive-like behaviors (TST), or sensorimotor gating (PPI) compared to controls (fig. S10). This null phenotype across multiple behavioral domains suggests that PFC-specific overexpression of either *DOC2A* isoform does not recapitulate the schizophrenia-related behavioral abnormalities observed following hippocampal overexpression. The notable contrast between these negative results and our hippocampal overexpression findings highlights the neuroanatomical specificity of *DOC2A*$^{\Delta Val217\text{-}Pro218}$'s behavioral effects, with hippocampal but not PFC expression sufficient to induce schizophrenia-relevant phenotypes in adult mice.

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DOC2A$^{\Delta Val217\text{-}Pro218}$ regulates excitatory synaptic transmission

DOC2A has been implicated in modulating spontaneous glutamate release (*39*); however, the role of *DOC2A*$^{\Delta Val217\text{-}Pro218}$ in this process remains elusive. To investigate the functional changes resulting from *DOC2A*$^{\Delta Val217\text{-}Pro218}$, we infected primary hippocampal neurons with *DOC2A*$^{\Delta Val217\text{-}Pro218}$ or *DOC2A*$^{Full\text{-}Length}$ AAV virus and assessed alterations in miniature synaptic phenotypes. The AAV constructs infected both excitatory and inhibitory neurons. To specifically record from excitatory neurons, we used CaMKIIα (calcium- and calmodulin-dependent protein kinase IIα)–driven GFP (green fluorescent protein) labeling for targeted electrophysiological recordings. Our results demonstrate that *DOC2A*$^{\Delta Val217\text{-}Pro218}$ overexpression significantly increased both the frequency and amplitude of miniature excitatory postsynaptic currents (mEPSCs). Parallel experiments revealed that *DOC2A*$^{Full\text{-}Length}$ overexpression exerted comparable effects on mEPSC frequency and amplitude, consistent with previous studies (amplitude: control, 11.90 ± 0.67; *DOC2A*$^{Full\text{-}Length}$, 14.44 ± 0.82; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 14.18 ± 0.75; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.019$, $t = 2.397$, df = 88; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.032$, $t = 2.186$, df = 88; frequency: control, 1.98 ± 0.39; *DOC2A*$^{Full\text{-}Length}$, 3.60 ± 0.69; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 3.59 ± 0.56; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.043$, $t = 2.056$, df = 88; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.04$, $t = 2.082$, df = 88; multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 5, A to C). In contrast, neither *DOC2A*$^{\Delta Val217\text{-}Pro218}$ nor *DOC2A*$^{Full\text{-}Length}$ overexpression altered the frequency or amplitude of miniature inhibitory postsynaptic currents (mIPSCs; Fig. 5, D to F). To explore the molecular mechanisms underlying these observations, we quantified the protein levels of synaptophysin and vesicle-associated membrane protein 2 (VAMP-2), key components of the presynaptic vesicle release machinery, in hippocampal neurons following *DOC2A* overexpression. Notably, synaptophysin expression was significantly elevated in neurons overexpressing either *DOC2A*$^{\Delta Val217\text{-}Pro218}$ or *DOC2A*$^{Full\text{-}Length}$ (relative synaptophysin level: control, 1 ± 0.02; *DOC2A*$^{Full\text{-}Length}$, 1.245 ± 0.02; *DOC2A*$^{\Delta Val217\text{-}Pro218}$, 1.16 ± 0.04; *DOC2A*$^{Full\text{-}Length}$ versus control, $P = 0.0006$, $t = 6.555$, df = 6; *DOC2A*$^{\Delta Val217\text{-}Pro218}$ versus control, $P = 0.0056$, $t = 4.217$, df = 6, multiple comparisons using Fisher's LSD in a one-way ANOVA; Fig. 5H), whereas VAMP-2 levels remained unchanged (Fig. 5I). Given the relevance of inhibitory neuron markers in schizophrenia pathology (*40–43*), we also measured the expression of glutamic acid decarboxylase 65 (GAD65) and GAD67. No significant differences in GAD65 or GAD67 protein levels were observed compared to the control group (Fig. 5, J and K). Collectively, these findings suggest that *DOC2A*$^{\Delta Val217\text{-}Pro218}$ overexpression enhances excitatory synaptic transmission, potentially contributing to behavioral abnormalities while leaving inhibitory synaptic transmission unaffected.

The alternatively spliced isoform *DOC2A*$^{\Delta Val217\text{-}Pro218}$ has distinct synaptic functions compared with *DOC2A*$^{Full\text{-}Length}$

Our integrated behavioral analyses reveal functional divergence between *DOC2A* isoforms. As we confirmed that both isoforms encode proteins, we found that DOC2A$^{\Delta Val217\text{-}Pro218}$ lacks two amino acids (Val^{217}-Pro^{218}) in a solvent-exposed loop region compared with DOC2A$^{Full\text{-}Length}$. Structural modeling indicated that this deletion maintains the overall protein architecture while shortening the surface loop (fig. S11). Given the hydrophobic nature of the deleted residues, we hypothesize that this modification may enhance structural stability through reduced solvent accessibility and improved hydrophobic core packing. Multiple subcellular localization studies, including synaptosomal fractionation, immunoelectron microscopy, and immunofluorescence, revealed that both DOC2A isoforms are similarly distributed across presynaptic and postsynaptic compartments (figs. S12 and S13), indicating that their functional distinctions

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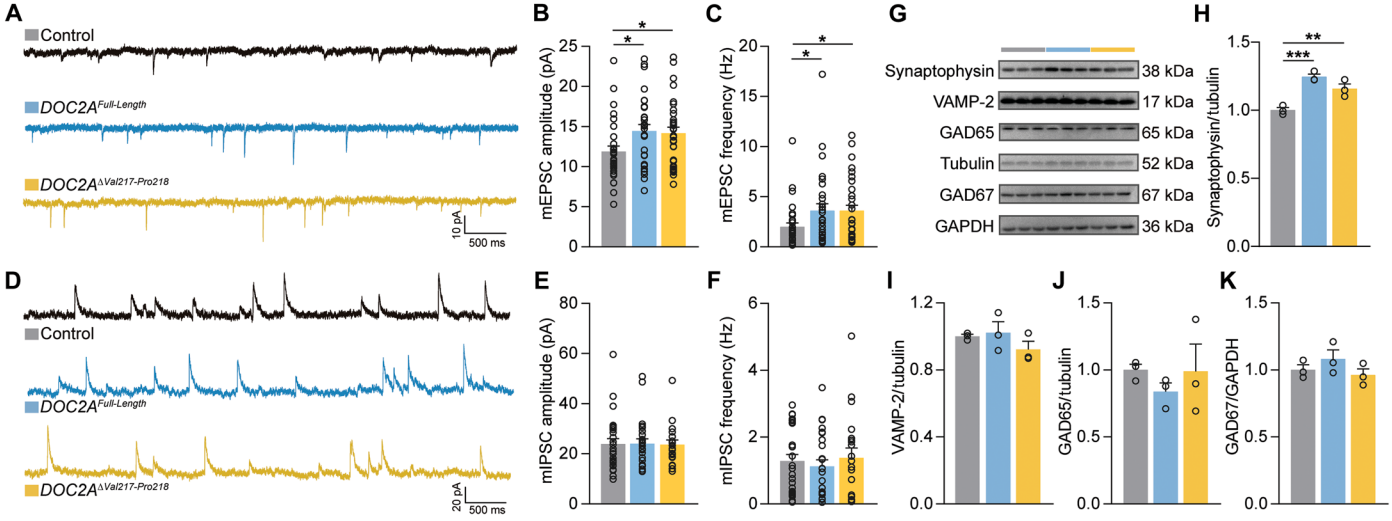

Fig. 5. *DOC2A*$^{\Delta Val217\text{-}Pro218}$ overexpression alter the excitatory synaptic transmission. (**A**) Representative traces of mEPSC in primary hippocampal neuron infected with control (empty vector), *DOC2A*$^{Full\text{-}Length}$, or *DOC2A*$^{\Delta Val217\text{-}Pro218}$ AAV. (**B** and **C**) Quantification of mEPSC amplitude and frequency. $N \geq 29$ neurons. (**D**) Representative traces of mIPSC in in primary hippocampal neuron infected with control, *DOC2A*$^{Full\text{-}Length}$, or *DOC2A*$^{\Delta Val217\text{-}Pro218}$ AAV. (**E** and **F**) Quantification of mIPSC amplitude and frequency. $N \geq 20$ neurons. (**G**) Representative gels of Western blot. (**H** to **K**) Quantification of synaptophysin, VAMP-2, GAD65, and GAD67 expression in hippocampus tissue overexpressing control, *DOC2A*$^{Full\text{-}Length}$, or *DOC2A*$^{\Delta Val217\text{-}Pro218}$. $N = 3$ in each group. Full Western blotting images are shown in fig. S16. Data are presented as the means ± SEM; $^{*}P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$. Error bars indicate the SEM.

likely arise from differential molecular interactions rather than distinct localization. These findings provide structural insights into how alternative splicing of *DOC2A* may generate functionally distinct isoforms that differentially regulate synaptic function and behavior.

To systematically identify the isoform-specific protein interactome, we performed an unbiased immunoprecipitation–mass spectrometry (IP-MS) analysis of hippocampal tissues from adult male C57BL/6J mice overexpressing either $DOC2A^{\Delta Val217\text{-}Pro218}$ or $DOC2A^{Full\text{-}Length}$ in the dorsal hippocampus via AAV-mediated delivery. Following FLAG-based immunoprecipitation and stringent filtering [fold change >2.0 versus immunoglobulin G (IgG) controls], we identified 98 interacting proteins for $DOC2A^{\Delta Val217\text{-}Pro218}$ (tables S6 and S7) and 114 for $DOC2A^{Full\text{-}Length}$ (tables S8 and S9) across two independent experimental replicates.

Metascape-based Gene Ontology (GO) analysis (*44*) uncovered distinct functional signatures between the interactors of DOC2A isoforms; the background is set to all proteins detected in the IP-MS experiment (table S10). The $DOC2A^{\Delta Val217\text{-}Pro218}$ interactome showed pronounced enrichment for calmodulin binding, modulation of chemical synaptic transmission, and postsynaptic cytoskeleton (Fig. 6A and table S11), aligning with our electrophysiology findings. However, $DOC2A^{Full\text{-}Length}$ preferentially associated with myelin sheath, transport vesicle, and pyrophosphatase activity (Fig. 6B and table S12), which is consistent with DOC2A's established role in calcium-dependent exocytosis (*45*, *46*). To further explore putative biological roles of $DOC2A^{\Delta Val217\text{-}Pro218}$, a mature complex identification algorithm called Molecular Complex Detection (MCODE) in Metascape was used to automatically extract protein complexes embedded in a large network (*47*). MCODE cluster analysis further highlighted these differences. The $DOC2A^{\Delta Val217\text{-}Pro218}$ network contained three significant clusters (table S13), and the top two MCODE clusters are presented (Fig. 6C), including MCODE1 (myosin II complex, $\log_{10}P = -11.3$, score = 5.96), MCODE2 [ATP (adenosine 5′-triphosphate)–dependent protein folding chaperone, $\log_{10}P = -6.4$, score = 2.41]. $DOC2A^{Full\text{-}Length}$ formed five unique complexes (table S13), and the top two MCODE clusters are presented (Fig. 6D), containing MCODE1 (regulation of protein localization to the plasma membrane, $\log_{10}P = -6.0$, score = 2.7) and MCODE2 (protein folding chaperone complex, $\log_{10}P = -7.3$, score = 3.69). To further dissect functional divergence, we performed GO enrichment on the 37 $DOC2A^{\Delta Val217\text{-}Pro218}$ unique interactors and 53 $DOC2A^{Full\text{-}Length}$ unique partners (table S14). $DOC2A^{\Delta Val217\text{-}Pro218}$ unique interactors were linked to the myosin II complex, supramolecular fiber organization, and ankyrin binding, whereas $DOC2A^{Full\text{-}Length}$ unique partners were enriched in pyrophosphatase activity, myelin sheath, and ATP metabolic process (Fig. 6, E and F, and tables S15 and S16). This network divergence potentially contributed to their differential behavioral phenotypes. This dichotomy in molecular pathways, coupled with their common synaptic functions, suggests that alternative splicing generates DOC2A isoforms with both specialized and overlapping roles in synaptic architecture and function.

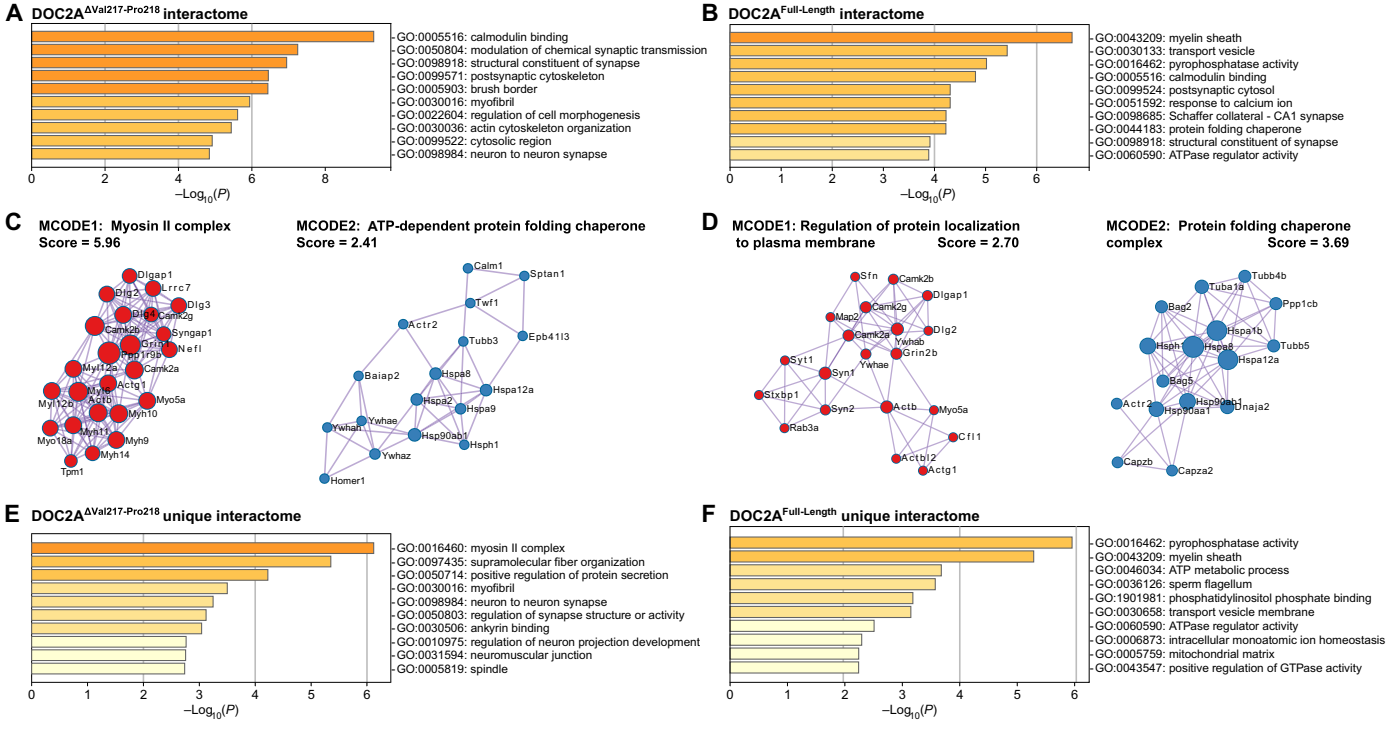

Fig. 6. Identification of the DOC2A interactome. Functional enrichment visualizations of the DOC2A interactome identified by IP-MS of mouse hippocampus using Metascape (https://metascape.org). (**A**) GO enrichment analysis of $DOC2A^{\Delta Val217\text{-}Pro218}$ interacting proteins. (**B**) GO enrichment analysis of $DOC2A^{Full\text{-}Length}$ interacting proteins. (**C**) Top two MCODE complexes identified in the $DOC2A^{\Delta Val217\text{-}Pro218}$ interactome. (**D**) Top two MCODE complexes identified in the $DOC2A^{Full\text{-}Length}$ interactome. (**E**) GO enrichment analysis of $DOC2A^{\Delta Val217\text{-}Pro218}$ unique interacting proteins. (**F**) GO enrichment analysis of $DOC2A^{Full\text{-}Length}$ unique interacting proteins. MCODE is a computational algorithm designed to identify densely connected clusters within PPI networks. The cluster score in MCODE reflects the quality and density of a detected cluster, calculated as the product of the cluster's density and the number of nodes in the cluster.

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DISCUSSION

Alternative splicing represents a crucial molecular mechanism bridging schizophrenia genetic risk with disease pathogenesis. While emerging evidence highlights significant enrichment of schizophrenia-associated SNPs in brain sQTLs, particularly within previously unannotated isoforms (*14*, *16*, *18*), our current understanding of brain transcriptome diversity remains incomplete (*15*, *48*). These uncharacterized splice isoforms often encode functionally divergent protein isoforms compared to their canonical counterparts; functional characterization of such previously unidentified isoforms therefore represents a critical opportunity to elucidate the molecular pathways underlying schizophrenia etiology and pathophysiology.

We identify rs3935873 as a functional variant modulating *DOC2A* alternative splicing within the schizophrenia-associated 16p11.2 locus. This genomic region harbors multiple GWAS-significant variants for schizophrenia and exhibits complex regulatory architecture, as evidenced by our previous finding of rs4420550-mediated chromatin looping (*4*). The synonymous variant rs3935873 (in moderate LD with rs4420550, $r^2 = 0.66$) creates a cryptic splice donor site through its risk G-allele, illustrating the pleiotropic functional consequences of variants within a single GWAS locus (*49*). While rs3935873 also shows eQTL associations with *INO80E* mRNA expression in BrainSeq DLPFC and hippocampal datasets (fig. S14), these likely reflect linkage with rs4420550 rather than direct regulatory effects, as supported by the absence of rs3935873 in DNA enhancer regions and lack of chromatin interactions in 3DIV (www.3div.kr) Hi-C data (*50*). These findings underscore the need to investigate cooperative mechanisms among multiple risk variants at 16p11.2 to fully elucidate their collective impact on schizophrenia pathogenesis.

Mechanistic investigation of rs3935873 revealed its specific association with the previously unidentified alternatively spliced $DOC2A^{\Delta Val217\text{-}Pro218}$ isoform rather than the canonical $DOC2A^{Full\text{-}Length}$ transcript, and this SNP is not polymorphic in mice but corresponds to a conserved G-allele. Moreover, analysis of RNA-seq data from mouse hippocampal and cortical tissues revealed no presence of $DOC2A^{\Delta Val217\text{-}Pro218}$ isoform, suggesting that this splicing event is likely absent in mice and may be a human-specific alternative splicing mechanism. Notably, the sQTL effects we observed exhibited substantially greater statistical significance than the schizophrenia GWAS signal at this locus despite the GWAS sample size being ~1000-fold larger. This pronounced discrepancy underscores how genetic variants may exert stronger effects on molecular phenotypes than on disease diagnosis per se, consistent with the endophenotype concept in psychiatric genetics (*51*). Our findings support the paradigm that molecular-level alterations, particularly in gene expression and splicing patterns, represent more direct and penetrant manifestations of genetic risk that operate through schizophrenia-relevant biological pathways before clinical symptom emergence. Furthermore, human pluripotent stem cell–derived organoids were used to model human brain development, including synapse formation and neural circuit organization, which have advanced our understanding of neurodevelopmental processes linked to schizophrenia (*52*). The integration of CRISPR-based gene editing with hippocampal organoids could, in future studies, enable a more precise investigation of how rs3935873 variants and $DOC2A^{\Delta Val217\text{-}Pro218}$ expression influence neurodevelopment in schizophrenia.

Some studies have reported hippocampal volume reduction in patients with schizophrenia. However, our neuroimaging analysis identified a significant association between the rs3935873 variant and increased hippocampal volume, with the schizophrenia-risk G-allele predicting larger volumes. This apparent contradiction suggests that rs3935873 may influence a distinct disease subtype, paralleling reports of other risk alleles (e.g., *ZNF804A* rs1344706) associated with hippocampal enlargement in patient subgroups (*53*) and in unaffected relatives with familial schizophrenia history (*54*). Notably, while adult hippocampal overexpression of $DOC2A^{\Delta Val217\text{-}Pro218}$ in our study did not alter gross morphology and hippocampus volume, it recapitulated key schizophrenia endophenotypes, including impaired prepulse inhibition. In contrast, this effect was specifically absent in both PFC $DOC2A^{\Delta Val217\text{-}Pro218}$-overexpressing and hippocampal $DOC2A^{Full\text{-}Length}$-overexpressing mice despite the conserved sQTL association observed across both brain regions. The regional specificity observed here likely stems from differences in local circuit integration (e.g., the hippocampus-ventral tegmental area axis, which mediates dopamine dysregulation and schizophrenia-relevant endophenotypes) (*55*, *56*), cell type–specific properties (e.g., the CA1 region of hippocampus is selectively vulnerable to a variety of metabolic and cytotoxic insults) (*57*), or distinct molecular contexts (*58*). Collectively, these findings implicate hippocampal-specific overexpression of $DOC2A^{\Delta Val217\text{-}Pro218}$ as sufficient to induce schizophrenia-relevant neural dysfunction and underscore that the phenotypic manifestation of genetic risks is profoundly shaped by the distinctive neurobiological context of the relevant brain region, highlighting the importance of isoform- and region-specific mechanisms in schizophrenia pathogenesis.

While *DOC2A* is conventionally recognized for its role in presynaptic glutamate release regulation (*45*, *46*), our patch-clamp recordings in cultured hippocampal neurons reveal that both $DOC2A^{\Delta Val217\text{-}Pro218}$ and $DOC2A^{Full\text{-}Length}$ isoforms, when overexpressed, significantly modulate mEPSC frequency and amplitude. However, through unbiased IP-MS analysis of the two isoforms, we found the $DOC2A^{\Delta Val217\text{-}Pro218}$ acquires distinct functions including the myosin II complex, supramolecular fiber organization, and ankyrin binding. These findings suggest that rs3935873-mediated alternative splicing may shift the DOC2A function from vesicle priming to cytoskeletal organization, potentially influencing synaptic architecture and plasticity.

While our study identifies $DOC2A^{\Delta Val217\text{-}Pro218}$ as a previously unidentified schizophrenia-associated isoform with distinct functional properties, several limitations warrant consideration. The behavioral phenotypes observed in hippocampal $DOC2A^{\Delta Val217\text{-}Pro218}$-overexpressing mice, while statistically significant, showed moderate effect sizes, a finding that necessitates cautious interpretation of their clinical relevance. Although comparative analyses revealed meaningful differences between $DOC2A^{\Delta Val217\text{-}Pro218}$ and $DOC2A^{Full\text{-}Length}$ in both behavioral and interacting proteins, potential variations in viral expression efficiency and ectopic protein localization due to overexpression across experimental groups may introduce subtle confounding effects. Furthermore, the current study provides only partial characterization of this isoform's spatial and temporal expression patterns, relying primarily on hippocampal-focused analyses. Critical questions remain regarding $DOC2A^{\Delta Val217\text{-}Pro218}$'s distribution across other brain regions and cell types, its potential developmental stage-specific effects, and its expression in peripheral tissues. For instance, the ventral hippocampus is implicated in schizophrenia pathophysiology, and the potential effects of $DOC2A^{\Delta Val217\text{-}Pro218}$ overexpression in this region require further exploration to clarify its role in disease mechanisms. In addition, it remains uncertain

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whether selective overexpression of $DOC2A^{\Delta Val217\text{-}Pro218}$ in excitatory or inhibitory neurons is sufficient to elicit schizophrenia-like behaviors. Future investigations incorporating comprehensive human transcriptomic datasets and spatiotemporal expression profiling will be essential to fully elucidate this isoform's role in schizophrenia pathogenesis.

Collectively, our findings emphasize the critical role of alternative splicing analysis in elucidating the molecular pathogenesis of schizophrenia. Through comprehensive characterization of a functionally distinct *DOC2A* isoform generated by a schizophrenia-associated genetic variant, we establish a critical mechanistic link between GWAS risk loci and synaptic dysfunction. This work not only advances our understanding of the genetic architecture of schizophrenia but also provides a framework for investigating how disease-associated splicing isoforms contribute to neuropsychiatric disorder pathophysiology. In addition, building upon these preliminary observations, we proposed some potential therapeutic directions worthy of further investigation: correcting splicing imbalances to favor the $DOC2A^{\Delta Val217\text{-}Pro218}$ isoform using approaches such as antisense oligonucleotides and stratifying patients by the rs3935873 genotype to guide personalized treatment, such as selecting antipsychotics known to modulate hippocampal excitability.

MATERIALS AND METHODS

Ethics

All experimental protocols involving murine models were conducted in full compliance with the ethical standards outlined by the National Advisory Committee for Laboratory Animal Research and were approved by the Animal Ethic Committee of Kunming Institute of Zoology (protocol no. IACUC-RE-2021-11-001), maintained under specific pathogen–free conditions, and acclimatized in standardized individually ventilated cages ($N = 5$ per cage) with controlled environmental parameters (ambient temperature of 22 ± 1°C, 55 ± 5% humidity, and 12:12-hour light-dark cycle with illumination commencing at 0800). Animals had unrestricted access to autoclaved rodent chow and sterile water throughout the experimental period, with environmental enrichment provided to minimize stress-related artifacts.

Human genomic analyses in this study were performed using aggregated, deidentified datasets obtained from publicly available repositories, including GWAS and sQTL resources. Given that no primary human subject data were generated, additional institutional review board approval was deemed unnecessary under prevailing ethical guidelines. All referenced primary datasets had previously undergone rigorous ethical review and received approval from their originating institutions' human research ethics committees, ensuring compliance with international standards for genomic research involving human-derived data.

Public GWAS statistics of schizophrenia

Genome-wide association statistics for schizophrenia were derived from the PGC3 European-ancestry meta-analysis, comprising 53,386 cases and 77,258 controls (dataset accession: https://doi.org/10.5281/zenodo.1251810) (*2*). Quality control procedures, genotyping methodology, and statistical analyses were performed as described in the original publication. To interrogate the 16p11.2 risk locus (chr16: 29,670,000 to 30,200,000; hg19), regional association signals were visualized using LocusZoom (http://locuszoom.sph.umich.edu/locuszoom/; accessed on 27 December 2025) (*59*), incorporating LD reference data from the 1000 Genomes Project European population.

Prediction analysis of alternative splicing using SpliceAI

To assess the potential impact of genetic variants on alternative splicing patterns, we used SpliceAI, a deep neural network–based predictive algorithm that computes position-dependent splice-altering probabilities (*23*). This framework generates four distinct delta scores (DS_AG, DS_AL, DS_DG, and DS_DL) representing the likelihood of acceptor gain, acceptor loss, donor gain, and donor loss events, respectively. The composite DS metric (range: 0 to 1), defined as the maximum value across these four scores, quantifies the probability of splicing disruption, with established thresholds of 0.2 (high recall), 0.5 (recommended), and 0.8 (high precision) used for functional interpretation. Furthermore, the algorithm provides delta position values that precisely localize predicted splicing alterations relative to the variant coordinates, where positive and negative delta position values indicate downstream and upstream effects, respectively, facilitating mechanistic interpretation of putative regulatory variants.

Junctional sQTL analysis in RNA-seq datasets

Junction-level sQTL analyses were performed using postmortem DLPFC RNA-seq data obtained from LIBD eQTL Browser Phase 1 (https://eqtl.brainseq.org/phase1/eqtl/), which integrates processed datasets from BrainSeq ($N = 412$) and CommonMind Consortium ($N = 285$). Similarly, junction-level sQTL data in the hippocampus were derived from LIBD eQTL Browser Phase 2 (http://eqtl.brainseq.org/phase2/eqtl/), which includes processed datasets from BrainSeq ($N = 340$). Following the pipeline described by Jaffe *et al.* (*18*, *58*), junction quantification was performed by filtering BAM files to retain only primary alignments (samtools view -bh -F 0 × 100) followed by junction extraction using regtools (version 0.1.0) to generate TopHat-compatible output. Normalized junction expression was quantified as reads per 80 million mapped to account for sequencing depth variation. sQTL identification was conducted using MatrixEQTL (*60*) with covariate adjustment for population stratification (three multidimensional scaling components for BrainSeq; five for CommonMind Consortium), sex, diagnostic status, and technical variation (26 expression principal components for BrainSeq; 10 for CommonMind Consortium), ensuring robust detection of splicing-associated genetic variants while controlling for potential confounding factors.

DOC2A minigene constructs

The splicing regulatory function of *DOC2A* intron 5-exon 8 was investigated using a minigene reporter system constructed in the Exontrap pET01 vector (MoBiTec GmbH). Genomic fragments encompassing the target exons with native flanking intronic sequences were PCR amplified using Bam HI/Not I–tagged primers (forward: 5′-CGCGGATCCGGACAGAGGAATGGGCAAAT-3′; reverse: 5′-AAATATGCGGCCGCCAAGATGCCTACCAGCAGTCC-3′) and directionally cloned into the multiple cloning site. The rs3935873-G → A variant was engineered using overlap extension PCR-mediated site-directed mutagenesis, creating isogenic constructs differing solely at this nucleotide position (verified by bidirectional Sanger sequencing). The pET01 vector's SV40 promoter–driven expression system, featuring chimeric 5′/3′ exons flanking a modified intronic sequence, enables quantitative assessment of alternative splicing patterns for inserted genomic fragments through constitutive transcription and processing.

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In vitro functional splicing assay

The HEK293T cells were originally obtained from the Kunming Cell Bank (Kunming Institute of Zoology), while the human neuroblast (SH-SY5Y) cells and the mouse neuroblast (Neuro-2a) cells were sourced from American Type Culture Collection. All cell lines were cultured in Dulbecco's modified Eagle's medium (Gibco) supplemented with 10% fetal bovine serum (Royacel), 1 mM sodium pyruvate, 1× nonessential amino acids, and 1% penicillin-streptomycin [penicillin (100 U/ml) and streptomycin (100 μg/ml)] and maintained at 37°C in a humidified 5% CO_2 atmosphere. Cell line authenticity was routinely verified by morphological inspection, and mycoplasma contamination was excluded through periodic PCR testing.

For splicing analysis, six-well plates were transfected (Lipofectamine 3000) with isogenic pET01 constructs (rs3935873-G vs A), with biological triplicates across three independent experiments. Total RNA isolated by TRIzol (Ambion) was reverse transcribed into cDNA (Thermo Fisher Scientific). Specific reverse primers (*DOC2A*$^{Full\text{-}Length}$: 5′-GAAGAGGGGGACGCCAGCGGGAC[C/T]T-3′; *DOC2A*$^{\Delta Val217\text{-}Pro218}$: 5′-GAAGAGGGGGACGCCAG[C/T]T-3′) paired with a common forward primer (5′-TAAGCTAATTCCTGCAGCCCG-3′) enabled isoform-discriminating amplification of exogenous transcripts. Expression quantification was normalized against the constitutive vector exon (forward: 5′-GGATGGGGGTGTCTACGGTG-3′; reverse: 5′-ACCACAAAGGTGCTGTTTGAC-3′) to control transfection variability. Amplification efficiency exceeded 90% for all primer pairs. Reaction specificity was confirmed through melt curve analysis and Sanger sequencing of agarose-purified products (2% gel).

GWAS of brain imaging phenotypes

The neuroimaging associations of rs3935873 were extracted from ENIGMA-VIS (https://enigma-brain.org/enigmavis/) (*28*), a comprehensive repository of GWAS data on structural neuroimaging phenotypes from European-ancestry populations generated by the ENIGMA consortium (*29*, *61*). This resource incorporates two major GWAS meta-analyses: (i) ENIGMA 5 examined intracranial volume and nine subcortical structures (brainstem, caudate, putamen, hippocampus, globus pallidus, thalamus, nucleus accumbens, amygdala, and ventral diencephalon) across 74,898 participants (*29*), while (ii) ENIGMA 3 analyzed cortical surface area and thickness measures (both global and across 34 functionally specialized regions) in 51,665 individuals (*61*).

DOC2A$^{Full\text{-}Length}$ and *DOC2A*$^{\Delta Val217\text{-}Pro218}$ overexpression plasmid construction, AAV package, and purification

To generate AAV constructs, we cloned the coding sequences of *DOC2A*$^{Full\text{-}Length}$ and *DOC2A*$^{\Delta Val217\text{-}Pro218}$ into the pAAV-CAG-tdTomato backbone (Addgene, no. 59462), yielding pAAV-CAG-*DOC2A*$^{Full\text{-}Length}$-P2A-mCherry and pAAV-CAG-*DOC2A*$^{\Delta Val217\text{-}Pro218}$-P2A-mCherry. The protein expression of DOC2A$^{Full\text{-}Length}$ and DOC2A$^{\Delta Val217\text{-}Pro218}$ was detected by Western blotting, with no significant differences in expression efficiency between the two isoforms (fig. S15). Recombinant AAV-DJ particles were produced via triple transfection in HEK293T cells, purified using iodixanol gradient centrifugation (lysis buffer: 150 mM NaCl and 20 mM tris-HCl, pH 8.0), and quantified by SYBR Green-based RT-qPCR targeting the ITR region (Addgene protocol). Viral titer determination [GC (genome copies)/ml] and production protocols followed our established methodology (*62*).

Stereotaxic surgery and microinjection

Bilateral dorsal hippocampal CA1 overexpression of *DOC2A* isoforms was performed in 7- to 8-week-old C57BL/6J mice through stereotaxic microinjection. Following isoflurane anesthesia (3% induction and 1% maintenance), AAV vectors (5×10^{12} GC/ml, 0.25 μl) were infused at coordinates relative to bregma [AP (anteroposterior): −2.06 mm; ML (mediolateral): ±1.45 mm; DV (dorsoventral): −1.45 and −1.65 mm from bregma] using a Hamilton syringe, with a 7-min dwell time to ensure complete diffusion. Postsurgical management included analgesic/anti-inflammatory administration and daily health monitoring, followed by a 4-week recovery period in home cages to allow for stable gene expression before behavioral testing.

Micro-CT imaging for hippocampal volume measurement in mice

Mice were transcardially perfused with 4% paraformaldehyde (PFA), and brains were postfixed overnight in PFA at 4°C to preserve tissue integrity. Subsequently, the brains were transferred to phosphate-buffered saline (PBS), and the hippocampus was gently dissected for further processing. Three-dimensional data were acquired for specimen by high-resolution in vivo and in vitro micro–computed tomography (micro-CT; NEMO II, PINGSENG Healthcare Inc., China). The specimen was placed between the x-ray source and the complementary metal-oxide semiconductor detector. During scanning, they were rotated 360° along the vertical long axis and kept within the scanning area, with 10,000 projections by the complementary metal-oxide semiconductor detector transmitted to a computer equipped with image processing software. The specimen was scanned at a voltage of 90 kV and a 0.05 mA current, and the total scanning time was 10 min. After scanning, three-dimensional reconstruction was performed using Avatar version 3.0 software (PINGSENG Healthcare Inc., China). The iterative algorithm was used for reconstruction, the image resolution was 1000 by 1000 pixels, and the pixel size was 15 μm.

Behavioral tests

All behavioral experiments were performed during daylight hours (0900 to 1830) in a temperature- and humidity-controlled environment (23 ± 1°C and 45% relative humidity). Following a 7-day acclimation period involving daily gentle handling (5 min/day), subjects were evaluated using a standardized double-blinded experimental design. As the rs3935873 SNP did not demonstrate sex-specific effects, and to minimize variability because of hormonal fluctuations in female mice, only male mice were used for behavioral experiments. To minimize stress-induced artifacts, animals were habituated to the testing arena for ≥60 min preceding behavioral assessment. Experimental apparatuses were systematically sanitized with 70% ethanol between trials to eliminate residual olfactory cues. Locomotor activity and behavioral parameters were quantitatively analyzed using SuperMaze, SuperTST, and Visustartle platforms (XinRuan Information Technology, Shanghai). To control for potential confounds associated with cyclic hormonal fluctuations, the study exclusively used age-matched male C57BL/6J mice in all experimental cohorts. The experiments were randomized and double blinded.

Open field test

Exploratory behavior was quantified in a 40 by 40 by 40 cm open field arena partitioned into central (20 by 20 cm) and peripheral zones. Each trial commenced with mice positioned in a corner

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facing the wall, allowing 5 min of unrestricted exploration. The distance traveled and time spent in different areas were recorded and analyzed.

Elevated zero maze (EZM)

The EZM consisted of a circular runway (45-cm outer diameter and 33-cm inner diameter) with alternating open (unprotected) and enclosed quadrants. Each trial initiated with subject placement at the open-enclosed transition zone, oriented toward the open quadrant, for a 5-min observation period. Anxiety-like behavior was quantified via the total distance, the percentage of the distance moved in the open and closed arms.

Novel object recognition

The open field apparatus [40 cm by 40 cm by 40 cm, length (L) × width (W) × height (H)] was used for this experiment. On day 1, mice underwent 5 min of spontaneous exploration without objects. On day 2, they were exposed to two identical cubic objects (wooden, 2 cm^3, orange) for 10 min (identical object session). On day 3, 24 hours postacquisition, one of the objects was replaced with a novel prism (wooden, 2 cm by 2 cm by 3 cm, green), and mice were allowed to explore both objects for 10 min (novel object session). To account for potential spatial bias, the positions of the familiar and novel objects were alternated between trials during the novel object session. Exploration was defined as the nose pointing ≤2 cm from the object's surface. The discrimination index was calculated as (time spent exploring the novel object − time spent exploring the familiar object)/(time spent exploring the novel object + time spent exploring the familiar object).

Y maze for spatial reference memory

A Y-shaped maze, consisting of three identical closed arms (35 cm by 5 cm by 15 cm, 120° interarm angles), was used for this test. The arms were labeled A, B, and C, with distinct geometric visual cues (circle versus triangle) at arm B/C terminals. The experiment consisted of two trials: a training session and a testing session, with a 24-hour intertrial interval. During the training session, mice were placed in arm A, with arm C (designated as the novel arm) closed off, and allowed to explore freely within arms A and B for 15 min. In the testing session, arm C was opened, and the mice were allowed to explore all three arms for 5 min. The time spent in the novel arm and the number of entries into the novel arm were recorded to assess the mice's spatial reference memory.

Y maze spontaneous alternation

The Y maze spontaneous alternation test was conducted using the aforementioned apparatus. Mice were positioned at the distal end of a designated start arm facing the central junction and allowed free exploration for 8 min. Arm entry sequences were analyzed using a strict criterion: consecutive entries into three distinct arms (e.g., ABC and BCA) qualified as a spontaneous alternation. The spontaneous alternation rate was calculated using the following formula: the number of spontaneous alternations/(total number of arm entries − 2).

Three-chamber social interaction test

A trichamber polypropylene arena (60 cm by 40 cm by 22 cm, $L \times W \times H$) with two transparent acrylic partitions was used to assess sociability and social novelty preference. Two grid enclosures were placed at the diagonal corners of the left and right chambers to house unfamiliar mice. The experiment followed a three-phase protocol. First, the test mouse was placed in the middle chamber and allowed to explore freely for 5 min with the enclosures empty. Afterward, a male C57BL/6J mouse (stranger 1, S1, age-matched) was placed in one enclosure, and the test mouse was allowed 10 min of exploration. In the final phase, a second unfamiliar mouse (stranger 2, S2) was placed in the other enclosure, and the test mouse interacted with both S1 and S2 for another 10 min. Exploration times for the empty enclosure, S1, and S2 (nose within 2 cm) were recorded. The social discrimination index was calculated as (time spent exploring S1 − time spent exploring the empty enclosure)/(time spent exploring S1 + time spent exploring the empty enclosure), and the social novelty discrimination index was calculated as (time spent exploring S2 − time spent exploring S1)/(time spent exploring S2 + time spent exploring S1).

Tail suspension test (TST)

TST quantifies depressive-like behavior by measuring stress-induced behavioral despair. Mice suspended by the distal tail (1 cm from the tip and 30-cm vertical displacement from the ground plane using adhesive fixation) exhibit escape attempts followed by progressive immobility, with depressive phenotypes showing prolonged immobility duration. The behavior was recorded using the SuperTST system over 6 min, and the immobility and climbing times during the final 4 min were analyzed.

Sucrose preference test (SPT)

The SPT involved an adaptation and experimental phase. Mice were housed individually and habituated to two bottles, one with 1% sucrose and the other with water, for 48 hours, with bottle positions swapped after 24 hours to prevent position bias. Before the experimental phase, mice underwent 24-hour water deprivation. Following this, they had access to both the sucrose solution and water for 12 hours, with the positions of the bottles switched once during the period. Conditions like no drinking, overdrinking, and no preference should be excluded from analysis (*63*). Sucrose consumption (%) = [sucrose intake (g)/total intake (g)] × 100%.

Prepulse inhibition

Sensorimotor gating was assessed via PPI using an acoustic startle reflex measurement system (VisuStartle, Shanghai XinRuan). Mice were placed in a plastic restrainer on a gravity-sensing platform within a soundproof chamber for 5 min of preadaptation, with a constant 62-dB background noise. In session 1, mice received 10 startle pulses (120 dB, 40 ms) at random intervals (average, 15 s). In session 2, five trial conditions were tested: prepulses of 74, 78, and 86 dB; a 120-dB startle pulse only; and combinations of the prepulses (74, 78, and 86 dB) with the 120-dB startle pulse. The 20-ms prepulse was presented 100 ms before the 40-ms startle pulse. Each trial was repeated 10 times in pseudorandom order with average intervals of 15 s. Startle responses were recorded, and prepulse inhibition was calculated as 100 − [(average startle response to prepulse)/(average startle response to startle pulse)] × 100 to assess the sensorimotor gating function.

Rotarod test

Motor coordination was evaluated using an accelerated rotarod paradigm following a 2-day acclimation period (10 rpm, 5 min/day). On the third day, mice were subjected to three test trials (5-min duration, 4- to 40-rpm acceleration) with randomized ramp profiles and 30-min intertrial intervals to minimize fatigue effects. Performance was quantified as the mean fall latency, defined as the duration of maintained balance before descent.

Statistical analyses of the behavioral data

Data were analyzed using GraphPad Prism 9.0 software and are presented as the means ± SEM. Group comparisons were performed using a one-way or two-way ANOVA. Significance was defined as $P < 0.05$ (#$P < 0.1$, *$P < 0.05$, **$P < 0.01$, and *$P < 0.001$). Sample sizes are indicated in the figure legends.

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Immunoelectron microscopy

Under ketamine anesthesia, mice were transcardially perfused with 15 ml of PBS (pH 7.4), followed by 20 ml of ice-cold 4% PFA. The dorsal hippocampus was fixed in 4% PFA/0.2% glutaraldehyde (0.1 M PBS, pH 7.4) at 4°C overnight, then washed four times in PBS for 15 min, and incubated in 0.1 M glycine/PBS at 4°C for 30 min. Following graded ethanol dehydration, the samples were embedded in LR White resin and ultraviolet polymerized at −20°C for 48 hours. Serial 90-nm thin sections were mounted on carbon-coated 200-mesh nickel grids. For immunolabeling, the grids were blocked with 1% bovine serum albumin, 0.05% Triton X-100, and 0.05% Tween 20 for 5 min at room temperature (RT) and then incubated with primary antibodies (FLAG, no. 8146S, Cell Signaling Technology, 1:5; DOC2A, no. PA5-31345, Invitrogen, 1:20) at 4°C overnight. Following PBS washes, secondary antibodies were incubated with 10-nm colloidal gold-labeled goat anti-mouse (no. G7652, Sigma-Aldrich, 1:10) and 5-nm colloidal gold-labeled goat anti-rabbit (no. G7277, Sigma-Aldrich, 1:20) for 2 hours at RT. The sections were then stained with 2% uranyl acetate and lead citrate and imaged using a JEM-1400 Plus TEM at 80 kV.

Primary neuron culture

Primary hippocampal neurons were isolated from embryonic day 16.5 to 17.5 C57BL/6 mice. Following rapid decapitation and hippocampus dissection, tissues were enzymatically digested for 20 min at 37°C using papain (Worthington Biochemical, no. LS003119) and deoxyribonuclease I (Sigma-Aldrich, D4263-1VL) and gently triturated to generate single-cell suspensions. Cells were plated on poly-D-lysine–coated [Corning, no. 354210; coated with poly-D-lysine (10 μg/ml) for ≥12 hours] round glass coverslips and maintained in Neurobasal medium (Gibco, 21103-049) supplemented with 2% B27 (Gibco, 17504-044), 1× GlutaMAX (Gibco, 35050-061), and 1% penicillin-streptomycin (Gibco, 15140122) at 37°C under 5% CO_2. The medium was partially replaced (50%) every 3 to 4 days.

Electrophysiological recording of primary hippocampal neurons

After 5 days in vitro (DIV5), neurons were infected with AAVs encoding an empty, $DOC2A^{Full\text{-}Length}$, or $DOC2A^{\Delta Val217\text{-}Pro218}$ construct. Whole-cell patch-clamp recordings were performed at DIV19 to allow sufficient transgene expression and neuronal maturation. Cells perfused with oxygenated artificial cerebrospinal fluid (125 mM NaCl, 2.5 mM KCl, 25 mM $NaHCO_3$, 1.25 mM NaH_2PO_4, 1 mM $MgCl_2$, and 1 mM $CaCl_2$, pH 7.4) were maintained at 32°C via a temperature-controlled perfusion system. Neurons exhibiting fluorescence (CaMKIIα-GFP) with pear-shaped soma were selected for recording. Miniature currents (mEPSCs/mIPSCs) were recorded in artificial cerebrospinal fluid containing 1 μM tetrodotoxin (Alomone Labs), and neurons were clamped at −70 mV to record mEPSCs and at 0 mV to record mIPSCs. Signals were acquired continuously for 5 min using MultiClamp 200B hardware (Molecular Devices), and a 3-min segment was analyzed offline with Clampfit software (Molecular Devices) to quantify the event frequency and amplitude.

Immunostaining

Primary hippocampal neurons (DIV7) infected with an empty, $DOC2A^{Full\text{-}Length}$, or $DOC2A^{\Delta Val217\text{-}Pro218}$ construct were processed for immunostaining upon viral expression. The supernatant was aspirated, and cells were washed twice with PBS at RT, fixed for 15 min in PBS containing 4% PFA and 4% sucrose (RT), and then washed three times with PBS (5 min per wash). Cells were blocked for 1 to 1.5 hours in PBS with 5% bovine serum albumin and 0.1% Triton X-100 (RT), incubated with a primary antibody diluted in blocking buffer overnight at 4°C, washed three times with PBS, and incubated for 1 hour (RT, dark) with a fluorophore-conjugated secondary antibody and DAPI (4′,6-diamidino-2-phenylindole) diluted in blocking buffer. After three final PBS washes, coverslips were air dried in the dark and mounted and imaged using a ZEISS LSM 910 with a 63× oil lens.

Empty, $DOC2A^{Full\text{-}Length}$, or $DOC2A^{\Delta Val217\text{-}Pro218}$ AAV was injected into the hippocampal CA1 region of mice. Four weeks postinjection, mice were transcardially perfused with 4% PFA; brains were postfixed, cryoprotected, and sectioned coronally at a 60-μm thickness using a vibrating microtome. Immunostaining followed the in vitro protocol with appropriately extended incubation and wash times. The brain slices were captured using an Olympus FV 4000 with a 60× oil lens.

Western blotting

Hippocampal tissue lysates were prepared in radioimmunoprecipitation assay buffer (supplemented with protease inhibitor cocktail, Roche) via 12,000*g* centrifugation (4°C, 10 min). The protein concentration was determined by BCA (bicinchoninic acid) assay (Thermo Fisher Scientific), and then the equal amounts of protein were denatured at 95°C. Samples were separated by 10% SDS–polyacrylamide gel electrophoresis gels and transferred to polyvinylidene fluoride membranes (EMD Millipore). The membranes were blocked with 5% skim milk and incubated with a primary antibody overnight at 4°C and horseradish peroxidase–conjugated secondary antibodies at RT. Chemiluminescent signals were captured on a Tanon 5200 system. The primary antibodies were as follows: FLAG (Cell Signaling Technology, no. 8146S; 1:1000) and DOC2A (Invitrogen, no. PA5-31345; 1:1000), GAD67 (Proteintech, no. 10408-1-AP; 1:5000), GAD65 (Cell Signaling Technology, no. 5843T; 1:1000), synaptophysin (Sigma-Aldrich, no. S5768; 1:500), VAMP-2 (Synaptic Systems, no. 104211; 1:2000), α-tubulin (Proteintech, no. 66031-1-Ig; 1:20,000), and GAPDH (glyceraldehyde-3-phosphate dehydrogenase; Proteintech, no. 10494-1-AP; 1:20,000). Blots were developed with horseradish peroxidase–conjugated secondary antibodies and quantified via densitometry using ImageJ software.

Structural modeling and analysis

Structural characterization of DOC2A isoforms was performed using in silico prediction approaches because of the lack of experimentally determined structures in public databases. The canonical amino acid sequence of $DOC2A^{Full\text{-}Length}$ (UniProt accession: Q14183) was retrieved from the UniProt database as input for structural prediction. AlphaFold3 calculations were performed with default parameters (*64*, *65*), and the model with the highest predicted TM-score from the five generated was selected for each group. Structural comparisons were conducted using PyMOL (PyMOL-3.1.3), with alignment performed using the align algorithm and default RMSD optimization. Visualizations were processed and annotated in Adobe Illustrator.

Preparation of subcellular fractions

Synaptosomal fractions were isolated from freshly dissected mouse brains through differential centrifugation in ice-cold sucrose buffer (0.32 M sucrose and 25 mM Hepes, pH 7.4) supplemented with

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protease inhibitor cocktail. Following initial low-speed centrifugation (700*g*, 10 min, 4°C) to remove nuclei and cellular debris, the supernatant was subjected to high-speed centrifugation (10,000*g*, 15 min, 4°C) to yield a crude synaptosomal fractions and cytosolic fraction. The synaptosomal fractions were resuspended in Hepes-buffered saline and treated with 1% Triton X-100 (4°C, 30 min) to solubilize non–postsynaptic density (PSD) components, followed by centrifugation (10,000*g*, 20 min, 4°C) to obtain a PSD-enriched pellet. Final PSD purification was achieved through sequential Triton X-100 washing and resuspension in 3% SDS–Hepes-buffered saline, yielding biochemically isolated PSD fractions suitable for downstream proteomic analysis.

IP-MS

More than three mice overexpressing *DOC2A*$^{Full\text{-}Length}$*-FLAG* and *DOC2A*$^{\Delta Val217\text{-}Pro218}$*-FLAG* were euthanized for IP-MS. The dorsal hippocampal tissues expressing fluorescent protein were homogenized in immunoprecipitation lysis buffer with protease inhibitors (no. 78430, Thermo Fisher Scientific) and incubated at 4°C for 30 min with gentle shaking. After centrifugation, supernatants were collected, and the protein concentration was determined using BCA Protein Assay Kits (no. 23225, Thermo Fisher Scientific). Equivalent aliquots were subjected to immunoprecipitation using either a control anti-IgG antibody (no. B900620, Proteintech) or anti-FLAG antibody (no. 8146S, Cell Signaling Technology). Samples were incubated overnight at 4°C. Protein G magnetic beads were added to both samples and incubated for an additional 4 hours. Following four cold lysis buffer washes, captured proteins were eluted with 3× SDT buffer (300 mM tris and 12% SDS, pH 8.0, no dithiothreitol) at 95°C for 10 min, with eluates processed for subsequent liquid chromatography–tandem mass spectrometry analysis.

Enrichment analysis

GO functional enrichment analysis of DOC2A$^{Full\text{-}Length}$ and DOC2A$^{\Delta Val217\text{-}Pro218}$ interacting proteins was performed using Metascape Custom Analysis (https://metascape.org) with a statistical threshold of [Pathway & Process Enrichment, min overlap: 3, P cutoff: 0.01 and min enrichment: 1.5; Protein-protein Interaction Enrichment, databases: PHYSICAL_CORE, min network size: 3 and max network size: 500] (*44*). The background is set to all proteins detected in the IP-MS experiment. To delineate highly interconnected subnetworks within PPI networks, we used the MCODE algorithm integrated in Metascape. The MCODE cluster score, reflecting both the density and size of a cluster, was used to assess its biological significance. Clusters with scores greater than 2.0 were considered substantial (*66*), and all identified complete MCODE clusters are presented in table S13.

Supplementary Materials

The PDF file includes:
Figs. S1 to S16
Legends for tables S1 to S16

Other Supplementary Material for this manuscript includes the following:
Tables S1 to S16

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